

DEVOPS FUNDAMENTALS PROJECT

Linux • AWS EC2 • Apache • Bash • Cron • Git & GitHub

Final Project Report

A practical implementation of a Linux-based DevOps project covering cloud infrastructure, Linux administration, static website deployment, shell scripting, scheduled automation, monitoring, logging, backups, and version control.

Project Structure

DevOps-Fundamentals-Project/

■■■ README.md

■■■ index.html

■■■ health-check.sh

■■■ apache-monitor.sh

■■■ backup.sh

■■■ log-report.sh

■■■ log-report.txt

■■■ screenshots/

■■■ documentation/

Prepared for academic/project submission

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Chapter 1 – Introduction

DevOps brings development and operations activities together through automation, collaboration, repeatable processes, monitoring, and continuous improvement. This project demonstrates those fundamentals through a small but complete Linux server deployment.

An Ubuntu-based AWS EC2 instance is used as the server. Apache hosts a static HTML website, Bash scripts perform health checks, Apache monitoring, backups, and log reporting, Cron schedules recurring work, and Git/GitHub provides version control.

Chapter 2 – Project Background

The project demonstrates how a Linux server can be configured and maintained using common DevOps tools. Rather than treating server setup, application deployment, monitoring, automation, and source control as separate activities, the project combines them into one workflow.

Workflow: AWS EC2 provides compute, Ubuntu provides the operating system, Apache provides web serving, Bash provides automation, Cron provides scheduling, and Git/GitHub provides source control.

Chapter 3 – Project Objectives

The major objectives of the project were:

- Create and configure an AWS EC2 Linux server.
- Connect securely to the server using SSH.
- Practice essential Linux administration commands.
- Install and configure Apache.
- Deploy and verify a static HTML website.
- Create Bash scripts for health checks, monitoring, backups, and log reporting.
- Schedule automated tasks with Cron.
- Collect useful server and system information.
- Use Git and GitHub for version control.
- Organize screenshots, scripts, documentation, and source files for submission.
- Develop troubleshooting skills through real logs and command output.

Chapter 4 – Technologies Used

AWS EC2 – cloud virtual server used for the project.

Ubuntu Linux – operating system and command-line environment.

Apache HTTP Server – hosts and serves the static website.

Bash – automates operational tasks.

Cron – runs recurring scripts automatically.

Git – tracks project changes.

GitHub – stores the remote repository.

SSH – provides secure remote access.

HTML – provides static website content.

Chapter 5 – Project Architecture

The project follows a simple client-to-server architecture. A browser sends an HTTP request to the EC2 public IP. Apache receives the request and serves index.html. Administration is performed over SSH, while Bash and Cron handle automation.

Architecture flow

User / Browser → HTTP → AWS EC2 → Ubuntu Linux → Apache → index.html

The EC2 instance also contains Bash scripts, the Cron scheduler, logs/reports, and the Git working tree. This separates application serving, automation, scheduling, logging, and source control while keeping the project simple enough to understand and verify.

Chapter 6 – AWS EC2 Setup

An AWS EC2 instance running Ubuntu was used as the main project server. The instance was accessed remotely with SSH using a key pair. Network access was configured so SSH could be used for administration and HTTP could be used to reach the website.

Typical SSH command:

```
ssh -i ~/Downloads/Project1.pem ubuntu@<EC2-PUBLIC-IP>
```

The project directory was maintained at /home/ubuntu/DevOps-Fundamentals-Project. The private .pem key was kept outside the project repository for security.

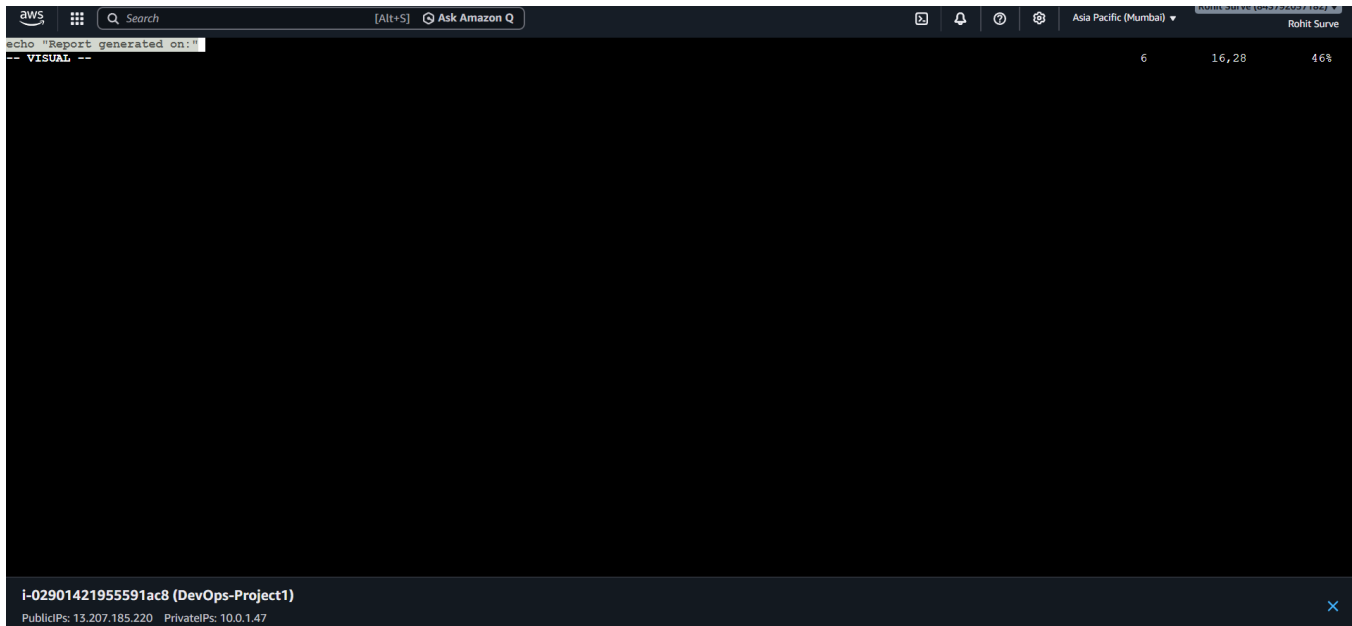


Figure 1. AWS EC2 environment showing the project instance and its public/private IP details.

Chapter 7 – Linux Commands

Linux command-line administration was used throughout the project:

pwd — current directory
 ls -la — list files
 cd <directory> — change directory
 mkdir <directory> — create directory
 touch <file> — create file
 cat <file> — display file
 vim <file> — edit file
 chmod +x script.sh — executable permission
 ps -ef — running processes
 df -h — disk usage
 free -h — memory
 uptime — uptime and load
 systemctl status apache2
 journalctl -u cron
 grep -i "error" /var/log/syslog

These commands supported configuration, monitoring, troubleshooting, and verification of the server.

```

aws [Search] [Alt+S] Asia Pacific (Mumbai) Rohit Surve (843792057182)
ubuntu@ip-10-0-1-47:~$ pwd
/home/ubuntu
ubuntu@ip-10-0-1-47:~$ ls
DevOps-Fundamentals-Project
ubuntu@ip-10-0-1-47:~$ cd ~/DevOps-Fundamentals-Project/
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ ./health-check.sh
=====
SERVER HEALTH CHECK
=====

1. CPU Usage
$Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st

2. Memory Usage
      total        used        free      shared  buff/cache   available
Mem:   908Mi       322Mi       265Mi        3.1Mi       431Mi       586Mi
Swap:   0B           0B           0B

3. Disk Usage
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            455M  0  455M  0% /dev/shm
tmpfs            162M  900K  161M  1% /run
efivarfs         128K  3.3K  120K  3% /sys/firmware/efi/efivars
none             1.0M  0  1.0M  0% /run/credentials/systemd-journald.service
tmpfs            455M  0  455M  0% /tmp
none            1.0M  0  1.0M  0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13  969M  163M  760M 18% /boot
/dev/nvme0n1p15  105M  6.3M  99M  7% /boot/efi
none            1.0M  0  1.0M  0% /run/credentials/systemd-networkd.service
none            1.0M  0  1.0M  0% /run/credentials/getty@tty1.service
none            1.0M  0  1.0M  0% /run/credentials/serial-getty@ttyS0.service
tmpfs            91M   8.0K  91M  1% /run/user/1000

4. Logged-in Users

5. Running Processes
UID          PID    PPID  C  STIME TTY          TIME CMD
i-02901421955591ac8 (DevOps-Project1)
PublicIPs: 13.206.237.125 PrivateIPs: 10.0.1.47
CloudShell Agent Toolkit for AWS Feedback Console Mobile App
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```

Figure 2. Health-check script output demonstrating Linux CPU, memory, disk, and process information.

Chapter 8 – Apache Web Server & Static Website Deployment

Apache HTTP Server was installed and configured to serve the project's static HTML page.

```
sudo apt update
sudo apt install apache2 -y
sudo systemctl enable apache2
sudo systemctl start apache2
sudo systemctl status apache2
```

The website content was prepared in index.html and deployed to the Apache document root. The deployment was verified through a web browser using the EC2 public address.

This demonstrates a basic release workflow: prepare source, deploy content, start or reload the service, and verify the application from a client.

```

• apache2.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
  Active: active (running) since Tue 2026-08-04 12:10:30 UTC; 41s ago
  Invocation: c63fefbcl4be41ef8937d898ea406841
  Docs: https://httpd.apache.org/docs/2.4/
  Main PID: 2734 (apache2)
  Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
  Tasks: 55 (limit: 627)
  Memory: 5.7M (peak: 5.7M)
  CPU: 61ms
  CGroup: /system.slice/apache2.service
          └─2734 /usr/sbin/apache2 -k start -DFOREGROUND
            └─2737 /usr/sbin/apache2 -k start -DFOREGROUND
              └─2738 /usr/sbin/apache2 -k start -DFOREGROUND

Aug 04 12:10:30 ip-10-0-1-47 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 04 12:10:30 ip-10-0-1-47 apachectl[2734]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 10.0.1.47. Set the 'ServerName' directive glo
Aug 04 12:10:30 ip-10-0-1-47 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ sudo systemctl status apache2
• apache2.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
  Active: active (running) since Tue 2026-08-04 12:10:30 UTC; 59s ago
  Invocation: c63fefbcl4be41ef8937d898ea406841
  Docs: https://httpd.apache.org/docs/2.4/
  Main PID: 2734 (apache2)
  Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
  Tasks: 55 (limit: 627)
  Memory: 5.7M (peak: 5.7M)
  CPU: 62ms
  CGroup: /system.slice/apache2.service
          └─2734 /usr/sbin/apache2 -k start -DFOREGROUND
            └─2737 /usr/sbin/apache2 -k start -DFOREGROUND
              └─2738 /usr/sbin/apache2 -k start -DFOREGROUND

Aug 04 12:10:30 ip-10-0-1-47 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 04 12:10:30 ip-10-0-1-47 apachectl[2734]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 10.0.1.47. Set the 'ServerName' directive glo
Aug 04 12:10:30 ip-10-0-1-47 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$

```

i-02901421955591ac8 (DevOps-Project1)

Public IPs: 13.206.237.125 Private IPs: 10.0.1.47

CloudShell Agent Toolkit for AWS Feedback Console Mobile App

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Figure 3. Apache service verification showing apache2 as active and enabled.

Chapter 9 – Bash Scripting

Bash scripts were created to automate repetitive administrative tasks. The main scripts are health-check.sh, apache-monitor.sh, backup.sh, and log-report.sh.

health-check.sh collects CPU, memory, disk, processes, users, and uptime information.

apache-monitor.sh checks Apache service status.

backup.sh creates repeatable project backups.

log-report.sh extracts useful warning and error information.

```
chmod +x health-check.sh
chmod +x apache-monitor.sh
chmod +x backup.sh
chmod +x log-report.sh
```

```
./health-check.sh
./apache-monitor.sh
./backup.sh
./log-report.sh
```

```
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ pwd
/home/ubuntu/DevOps-Fundamentals-Project
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ vim apache-monitor.sh
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ chmod +x apache-monitor.sh
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ ./apache-monitor.sh
=====
      APACHE SERVICE MONITOR
=====
Apache service is already running.
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ sudo systemctl stop apache2
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: inactive (dead) since Tue 2026-08-04 12:09:01 UTC; 5s ago
     Duration: 24min 53.859s
  Invocation: ea47845d0294494cac2513804661874f
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 579 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Process: 2690 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
    Main PID: 579 (code=exited, status=0/SUCCESS)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Mem peak: 11.1M
        CPU: 320ms
Aug 04 11:44:06 ip-10-0-1-47 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 04 11:44:07 ip-10-0-1-47 apachectl[579]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using ::1. Set the 'ServerName' directive globally to
Aug 04 11:44:07 ip-10-0-1-47 systemd[1]: Started apache2.service - The Apache HTTP Server.
Aug 04 12:09:01 ip-10-0-1-47 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Aug 04 12:09:01 ip-10-0-1-47 apachectl[2690]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 10.0.1.47. Set the 'ServerName' directive glo
Aug 04 12:09:01 ip-10-0-1-47 systemd[1]: apache2.service: Deactivated successfully.
Aug 04 12:09:01 ip-10-0-1-47 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
lines 1-20/20 (END)
```

Figure 4. Apache-monitor Bash script execution and service-status verification.

Chapter 10 – Cron Job Automation

Cron was configured to run the health-check script every minute. The configured entry was:

```
* * * * * /home/ubuntu/DevOps-Fundamentals-Project/health-check.sh >>
/home/ubuntu/health-report.txt
```

The five fields represent minute, hour, day of month, month, and day of week. The asterisks mean every possible value. The job was verified with `crontab -l`, while system logs confirmed repeated executions.

Useful verification commands:

```
grep CRON /var/log/syslog | tail -10
journalctl -u cron --since "5 minutes ago"
```

```
ubuntu@ip-10-0-1-47:~$ pwd
/home/ubuntu
ubuntu@ip-10-0-1-47:~$ system status cron
Command 'system' not found, did you mean:
  Command 'systemd' from deb systemd (235.1-6.3)
Try: sudo apt install <deb name>
ubuntu@ip-10-0-1-47:~$ systemctl status cron
● cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-08-06 12:07:53 UTC; 54min ago
     Invocation: s8eae9b6c5efc40ee8d495018116f1469
       DocDir: man/cron(8)
     Main PID: 599 (cron)
       Tasks: 1 (limit: 627)
     Memory: 516K (peak: 2.1M)
       CPU: 53ms
     CGroup: /system.slice/cron.service
            └─599 /usr/sbin/cron -f -S

Aug 06 12:07:53 ip-10-0-1-47 systemd[1]: Started cron.service - Regular background program processing daemon.
Aug 06 12:07:53 ip-10-0-1-47 (cron)[599]: cron.service: Referenced but unset environment variable evaluates to an empty string: EXTRA_OPTS
Aug 06 12:07:53 ip-10-0-1-47 cron[599]: (CRON) INFO (pidfile fd = 3)
Aug 06 12:07:53 ip-10-0-1-47 cron[599]: (CRON) INFO (Running @reboot jobs)
Aug 06 12:17:01 ip-10-0-1-47 CRON[1549]: pam_unix(cron:session): session opened for user root (uid=0) by root(uid=0)
Aug 06 12:17:01 ip-10-0-1-47 CRON[1571]: (root) CMD (cd / 44 run-parts --report /etc/cron.hourly)
Aug 06 12:17:02 ip-10-0-1-47 CRON[1569]: pam_unix(cron:session): session closed for user root
ubuntu@ip-10-0-1-47:~$
```

Figure 5. Cron service status and execution records on the Ubuntu EC2 server.

```
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m. every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
* * * * * /home/ubuntu/DevOps-Fundamentals-Project/health-check.sh >> /home/ubuntu/health-report.txt
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$
```

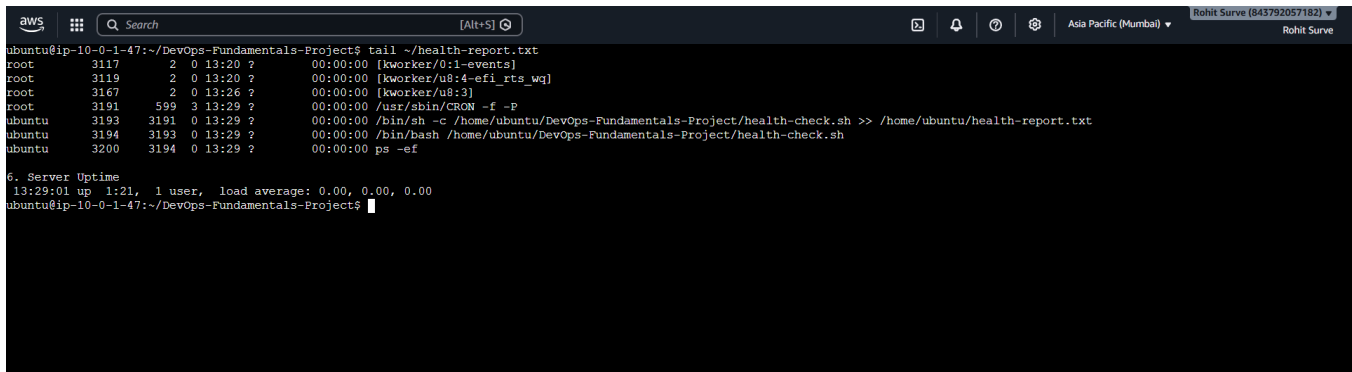
Figure 6. Cron-driven health-check execution visible in the server process output.

Chapter 11 – Server Health Monitoring

Health monitoring observes the state of the host and its services. The project checks processes, uptime, load, Apache status, and system logs.

The captured health report showed Cron launching health-check.sh and included an uptime line with a low load average. This provided evidence that the server was operating normally during testing.

Apache monitoring complements the general health check by checking the web service itself. Combining host and service checks makes troubleshooting faster because it distinguishes server-level issues from application or service-level issues.



```
aws | Search | [Alt+S] | Asia Pacific (Mumbai) | Rohit Surve (843792057182) | Rohit Surve
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$ tail ~/health-report.txt
root      3117      2   0 13:20 ?        00:00:00 [kworker/0:1-events]
root      3119      2   0 13:20 ?        00:00:00 [kworker/u8:4-cfi_rts_wq]
root      3167      2   0 13:26 ?        00:00:00 [kworker/u8:3]
root      3191    599   3 13:29 ?        00:00:00 /usr/sbin/CRON -f -P
ubuntu    3193    3191   0 13:29 ?        00:00:00 /bin/sh -c /home/ubuntu/DevOps-Fundamentals-Project/health-check.sh >> /home/ubuntu/health-report.txt
ubuntu    3194    3193   0 13:29 ?        00:00:00 /bin/bash /home/ubuntu/DevOps-Fundamentals-Project/health-check.sh
ubuntu    3200    3194   0 13:29 ?        00:00:00 ps -ef

6. Server Uptime
13:29:01 up 1:21, 1 user, load average: 0.00, 0.00, 0.00
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$
```

Figure 7. Automated health report showing Cron, health-check.sh, running processes, and server uptime.

Chapter 12 – Log Monitoring and Reporting

System logs were inspected with `grep`, `tail`, and `journalctl` to identify warnings, errors, and scheduled-job activity.

```
grep -i "error" /var/log/syslog | tail -20
grep -i "warn" /var/log/syslog | tail -10
journalctl -u cron --since "5 minutes ago"
grep CRON /var/log/syslog | tail -10
```

During testing, the logs contained Amazon SSM Agent messages indicating that EC2 instance-role credentials were not available. These messages were useful for diagnosis but were separate from the required Apache website deployment. The key lesson is to identify which service produced a warning and whether it affects the project's required functionality.

```
Report generated on:
=====
SYSTEM LOG REPORT
=====
latest 10 authentication logs:
2026-08-06T12:13:12.069428+00:00 ip-10-0-1-47 ec2-instance-connect[1247]: Providing ssh key from EC2 Instance Connect with fingerprint: SHA256:Bma7gLCcMF4cYv2wVzm9nmbYlYloGLgdnXmI+JnL8s,
request-id: 0b2c787d-f67a-453a-9d71-78065ef63796, for IAM principal: arn:aws:iam::843792057182:root
2026-08-06T12:13:12.676615+00:00 ip-10-0-1-47 ec2-instance-connect[1364]: Querying EC2 Instance Connect keys for matching fingerprint: SHA256:Bma7gLCcMF4cYv2wVzm9nmbYlYloGLgdnXmI+JnL8s
2026-08-06T12:13:12.728089+00:00 ip-10-0-1-47 ec2-instance-connect[1396]: Providing ssh key from EC2 Instance Connect with fingerprint: SHA256:Bma7gLCcMF4cYv2wVzm9nmbYlYloGLgdnXmI+JnL8s,
request-id: 0b2c787d-f67a-453a-9d71-78065ef63796, for IAM principal: arn:aws:iam::843792057182:root
2026-08-06T12:13:13.010033+00:00 ip-10-0-1-47 sshd-session[1106]: Accepted publickey for ubuntu from 13.233.177.3 port 22071 ssh2: ED25519 SHA256:Bma7gLCcMF4cYv2wVzm9nmbYlYloGLgdnXmI+JnL8s
2026-08-06T12:13:13.012620+00:00 ip-10-0-1-47 sshd-session[1106]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
2026-08-06T12:13:13.040631+00:00 ip-10-0-1-47 systemd-logind[627]: New session '1' of user 'ubuntu' with class 'user' and type 'tty'.
2026-08-06T12:13:13.075931+00:00 ip-10-0-1-47 (systemd): pam_unix(systemd-user:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
2026-08-06T12:13:13.077166+00:00 ip-10-0-1-47 systemd-logind[627]: New session '2' of user 'ubuntu' with class 'manager' and type 'unspecified'.
2026-08-06T12:17:01.995103+00:00 ip-10-0-1-47 CRON[1569]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-08-06T12:17:02.017165+00:00 ip-10-0-1-47 CRON[1569]: pam_unix(cron:session): session closed for user root

latest 10 system logs:
2026-08-06T12:36:46.976511+00:00 ip-10-0-1-47 systemd[1]: Reloading apache2.service - The Apache HTTP Server...
2026-08-06T12:36:47.054535+00:00 ip-10-0-1-47 apachectl[1826]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 10.0.1.47. Set the 'ServerName' directive globally to suppress this message
2026-08-06T12:36:47.069053+00:00 ip-10-0-1-47 systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
2026-08-06T12:36:47.077332+00:00 ip-10-0-1-47 systemd[1]: logrotate.service: Deactivated successfully.
2026-08-06T12:36:47.077634+00:00 ip-10-0-1-47 systemd[1]: Finished logrotate.service - Rotate log files.
2026-08-06T12:40:04.285134+00:00 ip-10-0-1-47 systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-08-06T12:40:04.310163+00:00 ip-10-0-1-47 systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-08-06T12:40:04.310531+00:00 ip-10-0-1-47 systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2026-08-06T12:43:06.261139+00:00 ip-10-0-1-47 irqbalance[612]: Cannot change IRQ 25 affinity: Permission denied
2026-08-06T12:43:06.264484+00:00 ip-10-0-1-47 irqbalance[612]: IRQ 25 affinity is now unmanaged

Report generated on:
Thu Aug 6 12:48:21 UTC 2026
ubuntu@ip-10-0-1-47:~/DevOps-Fundamentals-Project$

i-02901421955591ac8 (DevOps-Project1)
PublicIPs: 13.207.185.220 PrivateIPs: 10.0.1.47
CloudShell Agent Toolkit for AWS Feedback Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
30°C Mostly cloudy ENG IN 100% 06:18 PM 06-08-2026
```

Figure 8. System log output used to inspect server events, warnings, and errors.

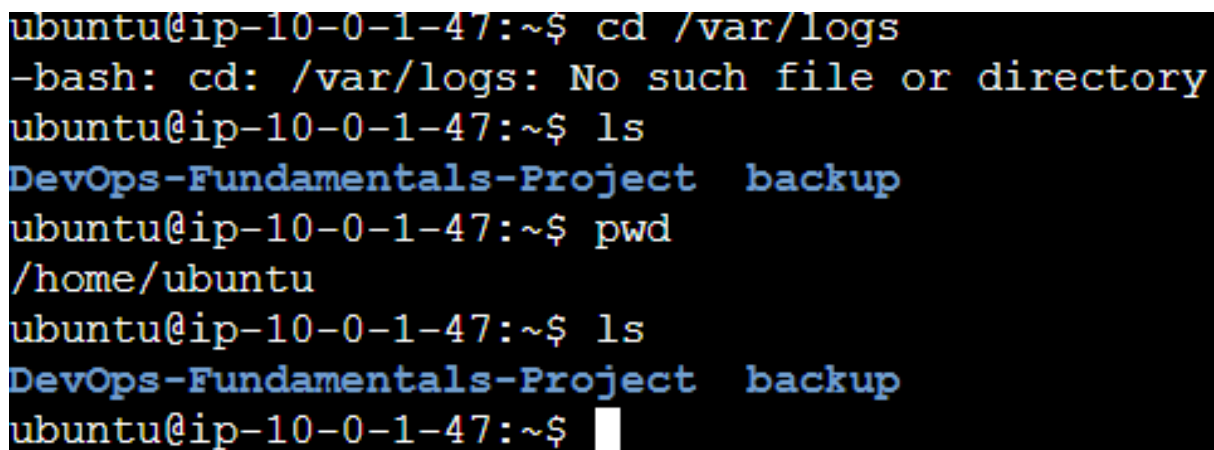
Chapter 13 – Backup Automation

Backups protect project files against accidental deletion or unwanted changes. The project includes backup.sh so the backup operation can be repeated consistently.

```
PROJECT="/home/ubuntu/DevOps-Fundamentals-Project"
BACKUP_DIR="/home/ubuntu/backups"
DATE=$(date +%Y-%m-%d_%H-%M-%S)

mkdir -p "$BACKUP_DIR"
tar -czf "$BACKUP_DIR/project_$DATE.tar.gz" "$PROJECT"
```

The key DevOps concept is repeatable, script-based operational automation. A script provides a consistent method of creating backups and reduces the chance of forgetting important files.

A terminal window screenshot showing a series of commands and their outputs. The user is at a shell prompt on a machine with IP 10.0.1.47. They attempt to change to /var/logs, which fails. Then they list the contents of their home directory, showing 'DevOps-Fundamentals-Project' and 'backup'. They then run 'pwd' to confirm they are in /home/ubuntu. Finally, they list the contents of their home directory again, showing the same two items. The terminal text is as follows:

```
ubuntu@ip-10-0-1-47:~$ cd /var/logs
-bash: cd: /var/logs: No such file or directory
ubuntu@ip-10-0-1-47:~$ ls
DevOps-Fundamentals-Project  backup
ubuntu@ip-10-0-1-47:~$ pwd
/home/ubuntu
ubuntu@ip-10-0-1-47:~$ ls
DevOps-Fundamentals-Project  backup
ubuntu@ip-10-0-1-47:~$
```

Figure 9. Backup directory and project-file operations on the EC2 server.

Chapter 14 – Git and GitHub Integration

Git was used to track project changes and GitHub was used as the remote repository.

```
git status
git add .
git commit -m "Update DevOps project"
git branch -M main
git push -u origin main
```

Git status was used to identify modified and untracked files before committing. The screenshot folder initially did not appear in GitHub because Git does not track empty directories. Once PNG files were added, the directory could be tracked. A filename typo was corrected from aws-steup.png to aws-setup.png.

```
rohit Survei@Rohit MINGW64 ~ (master)
$ pwd
/c/Users/saksh

rohit Survei@Rohit MINGW64 ~ (master)
$ cd ~/Desktop
bash: cd: /c/Users/saksh/Desktop: No such file or directory

rohit Survei@Rohit MINGW64 ~ (master)
$ ls
AppData/          'Local Settings'@  OneDrive/         Videos/
'Application Data'@  Music/             PrintHood@       'VirtualBox VMs'/
Contacts/         'My Documents'@    Recents          'WPS Cloud'/
Cookies/          NTUSER.DAT         'Saved Games'/   'WPS Cloud Files'/
Documents/        NTUSER.DAT[2ad838bc-efea-11ee-a54d-000d3a94eaal].TM.b1f  Searches/        WPSDrive/
Downloads/        NTUSER.DAT[2ad838bc-efea-11ee-a54d-000d3a94eaal].TM.Container00000000000000000001.regtrans-ms  SendTo          ntuser.dat.LOG1
Favorites/        NTUSER.DAT[2ad838bc-efea-11ee-a54d-000d3a94eaal].TM.Container00000000000000000002.regtrans-ms  'Start Menu'@   ntuser.dat.LOG2
Links/            NetHood@           Templates@        ntuser.ini

rohit Survei@Rohit MINGW64 ~ (master)
$ cd OneDrive
rohit Survei@Rohit MINGW64 ~/OneDrive (master)
$ ls
Attachments/ Desktop/ Documents/ 'Getting started with OneDrive.pdf' 'Personal Vault.Ink'* Pictures/ desktop.ini

rohit Survei@Rohit MINGW64 ~/OneDrive (master)
$ cd Desktop
rohit Survei@Rohit MINGW64 ~/OneDrive/Desktop (master)
$ ls
Cartforge/      'Opera Browser.Ink'*  'Razorpay Dashboard.Ink'*  'Zoom Workplace.Ink'*  devops/  referral_patients_2026-07-11.xlsx
'Google Meet.Ink'*  'Patient Data'/      'Rohit - Chrome.Ink'*     aws-steup.png          docu-A.pdf  website.png
'Grammarly.Ink'*   'Personal - Edge.Ink'*  'System Files'/          desktop.ini             github.png

rohit Survei@Rohit MINGW64 ~/OneDrive/Desktop (master)
$ scp -i ~/Downloads/Project1.pem *.png ubuntu@13.207.185.220:/home/ubuntu/DevOps-Fundamentals-Project/screenshots/
ssh: command not found

rohit Survei@Rohit MINGW64 ~/OneDrive/Desktop (master)
$ scp -i ~/Downloads/Project1.pem *.png ubuntu@13.207.185.220:/home/ubuntu/DevOps-Fundamentals-Project/screenshots/
The authenticity of host '13.207.185.220 (13.207.185.220)' can't be established.
ED25519 key fingerprint is: SHA256:0GHOH2b521KzEKbShA98Jawf4+oOute0788CJTEMKY
This host key is known by the following other names/addresses:
./ssh/known_hosts:4: 3.109.186.132
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.207.185.220' (ED25519) to the list of known hosts.
aws-setup.png 100% 312KB 4.6MB/s 00:00
github.png 100% 130KB 6.9MB/s 00:00
website.png 100% 86KB 4.2MB/s 00:00

rohit Survei@Rohit MINGW64 ~/OneDrive/Desktop (master)
$
```

Figure 10. Local Windows/OneDrive project files and SCP transfer of the project screenshots to EC2.

Chapter 15 – Testing and Verification

- EC2: confirmed successful SSH access.
- Linux: verified navigation, files, permissions, processes, and system information.
- Apache: checked service status and verified the website in a browser.
- Scripts: executed Bash scripts and reviewed output.
- Cron: verified the crontab entry and system execution records.
- Health report: checked health-report.txt for repeated output.
- Logs: used grep and journalctl for warnings, errors, and Cron records.
- Git: reviewed repository status and tracked project files.
- Project structure: verified scripts, website, screenshots, logs, README, and documentation.

Testing was performed incrementally so each component could be verified before moving to the next part of the project.

Chapter 16 – Troubleshooting and Lessons Learned

Several practical issues were resolved during the project. The Windows Desktop was located under OneDrive rather than the normal ~/Desktop path, so navigation through ~/OneDrive/Desktop was required.

SCP was used to copy PNG screenshots from Windows to the EC2 server. An initial command contained an accidental leading '\$', which caused a command-not-found error; removing the prompt symbol fixed it.

The screenshot directory was initially absent from GitHub because an empty directory is not tracked by Git. Adding image files solved this. The aws-steup.png typo was also corrected to aws-setup.png.

System logs showed SSM Agent credential warnings. These demonstrated real operational troubleshooting and reinforced the importance of distinguishing secondary service warnings from failures of the main Apache application.

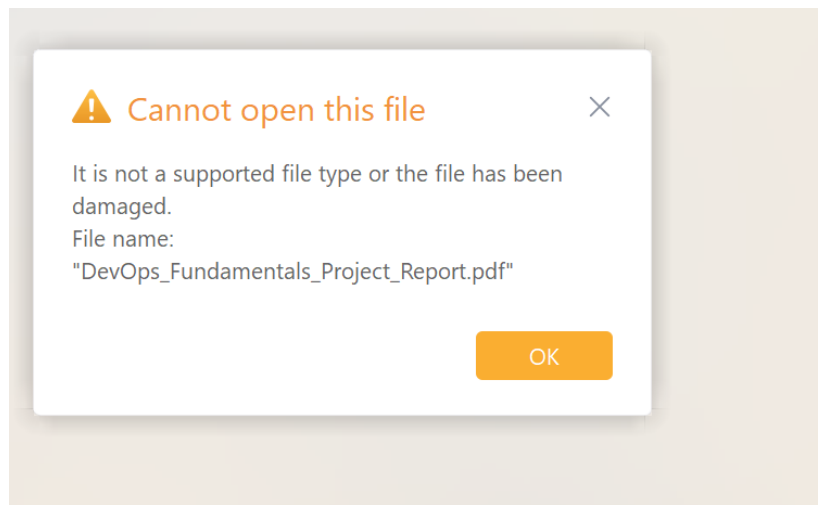


Figure 11. Evidence of troubleshooting the generated report file and validating the submission workflow.

Chapter 17 – Conclusion and Future Scope

The project combines cloud infrastructure, Linux administration, web deployment, shell scripting, scheduling, monitoring, logging, backups, and version control into one practical DevOps workflow.

The project demonstrates that DevOps is a combination of repeatable processes and tools rather than a single product. AWS EC2 provides infrastructure, Apache provides the service, Bash and Cron provide automation, and Git/GitHub provide version control.

Future enhancements: CI/CD with Jenkins or GitHub Actions, Docker, Terraform, centralized logging, HTTPS with a domain and TLS certificate, automated testing, and cloud monitoring with alarms.

Chapter 18 – Appendix: Final Project Structure and Evidence

The final project directory is organized as follows:

```
DevOps-Fundamentals-Project/
├── README.md
├── apache-monitor.sh
├── backup.sh
├── health-check.sh
├── index.html
├── log-report.sh
├── log-report.txt
├── documentation/
├── Project_Report.pdf
├── screenshots/
├── aws-setup.png
├── github.png
├── website.png
```

Evidence files: aws-setup.png documents AWS/EC2 setup; website.png documents the deployed static website; github.png documents the remote repository. The EC2 private key (.pem) is not part of the repository and must never be uploaded to GitHub.

```

Rohit Surve@Rohit MINGW64 ~ (master)
$ pwd
/c/Users/saksh

Rohit Surve@Rohit MINGW64 ~ (master)
$ cd ~/Desktop
bash: cd: /c/Users/saksh/Desktop: No such file or directory

Rohit Surve@Rohit MINGW64 ~ (master)
$ ls
Application Data/  Local Settings/  OneDrive/  Videos/
Application Data/  Music/           PrintHood/  VirtualBox VMs/
Contacts/         My Documents/    Recent/     WPS Cloud/
Cookies/          NTUSER.DAT       Searches/   WPS Cloud Files/
Documents/        NTUSER.DAT[2ad838bc-efea-11ee-a54d-000d3a94ea1].TM.b1f  SendTo/     WPS Drive/
Downloads/        NTUSER.DAT[2ad838bc-efea-11ee-a54d-000d3a94ea1].TM.Container00000000000000000001.regtrans-ms  Start Menu/  ntuser.dat.LOG1
Favorites/        NTUSER.DAT[2ad838bc-efea-11ee-a54d-000d3a94ea1].TM.Container00000000000000000002.regtrans-ms  Templates/   ntuser.dat.LOG2
Links/            NetHood/
$ cd OneDrive
Rohit Surve@Rohit MINGW64 ~OneDrive (master)
$ ls
Attachments/  Desktop/  Documents/  'Getting started with OneDrive.pdf'  'Personal Vault.lnk'  Pictures/  desktop.ini
Rohit Surve@Rohit MINGW64 ~OneDrive (master)
$ cd Desktop
Rohit Surve@Rohit MINGW64 ~OneDrive/Desktop (master)
$ ls
CartForge/  Google Meet.lnk  Opera Browser.lnk  Razorpay Dashboard.lnk  Zoom Workplace.lnk  devops/  referral_patients_2026-07-11.xlsx
Grammarly.lnk  Patient Data/  'Rohit - Chrome.lnk'  aws-setup.png  desktop.ini  github.png  website.png
Rohit Surve@Rohit MINGW64 ~OneDrive/Desktop (master)
$ scp -i ~/Downloads/Project1.pem *.png ubuntu@13.207.185.220:/home/ubuntu/DevOps-Fundamentals-Project/screenshots/
bash: scp: command not found

Rohit Surve@Rohit MINGW64 ~OneDrive/Desktop (master)
$ scp -i ~/Downloads/Project1.pem *.png ubuntu@13.207.185.220:/home/ubuntu/DevOps-Fundamentals-Project/screenshots/
The authenticity of host '13.207.185.220 (13.207.185.220)' can't be established.
ED25519 key fingerprint is: SHA256:0GhohZD52kZekbshX9sJbwfK4+oOuTe07BbcJTEMKY
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: 3.109.186.132
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.207.185.220' (ED25519) to the list of known hosts.
aws-setup.png                                100% 312KB  4.6MB/s  00:00
github.png                                  100% 130KB  6.9MB/s  00:00
website.png                                100% 86KB   4.2MB/s  00:00
Rohit Surve@Rohit MINGW64 ~OneDrive/Desktop (master)
$

```

Figure 12. Project evidence files and screenshot transfer workflow used to prepare the final repository.

Final Submission Checklist

- PDF opens successfully and contains all required chapters.
- README.md is present.
- index.html is present and the website has been verified.
- Bash scripts are present and executable where required.
- log-report.txt is present.
- screenshots/ contains aws-setup.png, github.png, and website.png.
- documentation/ contains the final report.
- Git status has been reviewed and intended files are committed/pushed.
- No .pem private key or secret credentials are included in GitHub.

Final Project Report

End of Report